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Mastery Project

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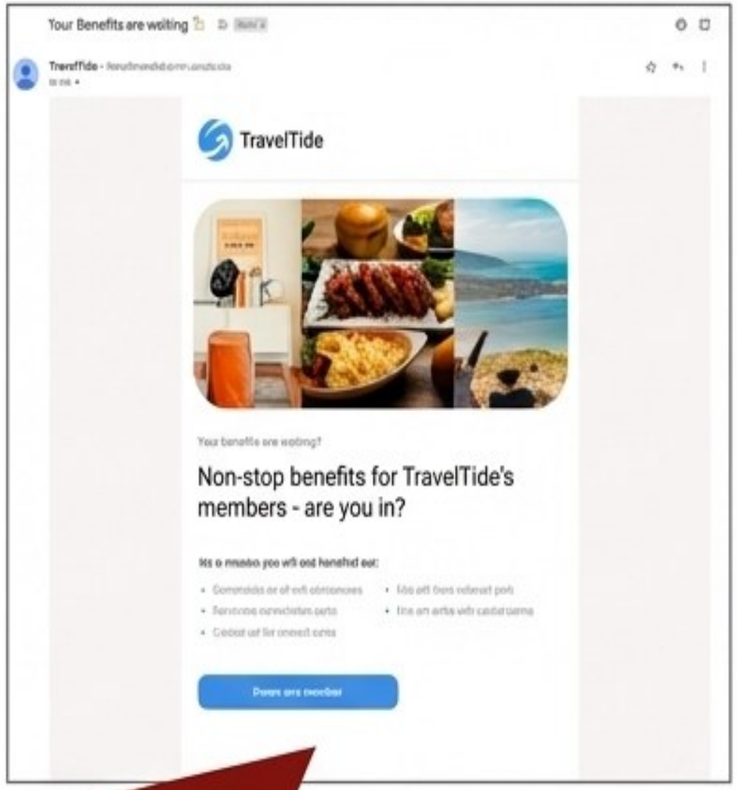
Precision Targeting for TravelTide Rewards

A Data-Driven Blueprint for Customer Segmentation and Perk Assignment

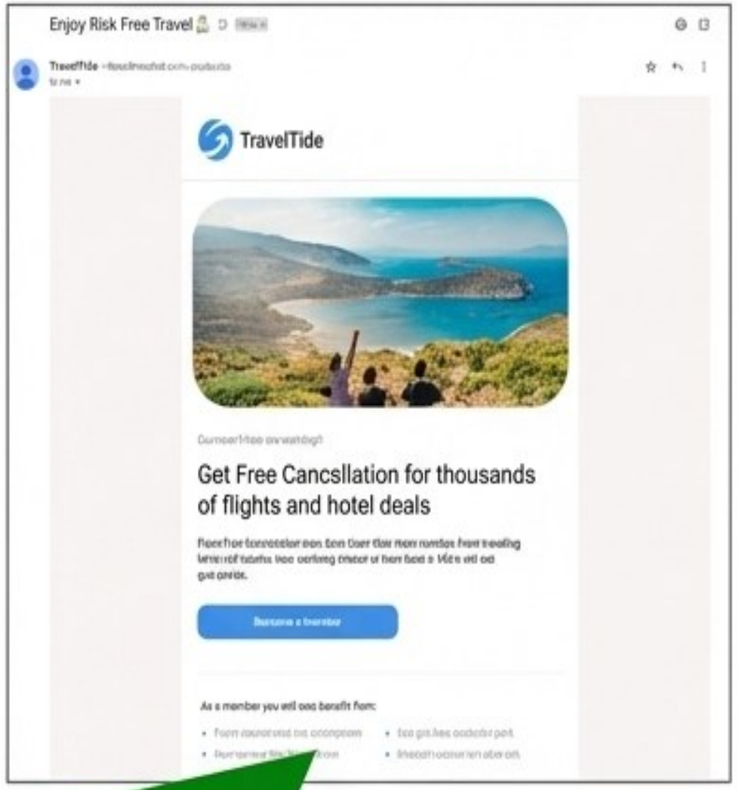
The Business Imperative: Moving from Mass Marketing to Personalization

The Problem: Generic Approach



A one-size-fits-all list of perks forces the customer to do the work, depressing engagement.

The Objective: Precision Targeting



Highlighting the single most relevant perk directly captures attention and drives reward program sign-ups.

Our mandate: Analyze behavioral data to assign every eligible customer to their highest-affinity reward perk.

From Raw Sessions to Actionable Customer Profiles

Step 1

Raw Data Aggregation

Combined session logs, flight history, and hotel data into one unified analytical view.

Step 2

Active Intent Refinement

Focused the analysis on an active cohort: users with 7+ sessions since January 2023.

Step 3

User-Level Profiles

Aggregated session behavior into multi-dimensional user profiles for segmentation.

Data Scope

- **49,211 filtered sessions**
- **5,998 users**
- **Active cohort: 7+ sessions since January 2023**

Session data were validated and cleaned before user-level aggregation and feature creation.

Key Behavioral Signals at the Session Level

Hotel discounts were associated with higher conversion (37.26%) than flight discounts, which showed almost no effect.

Page-click activity served as a strong indicator of booking intent. Sessions with bookings averaged 29.6 clicks versus 11.4 for non-booking sessions.

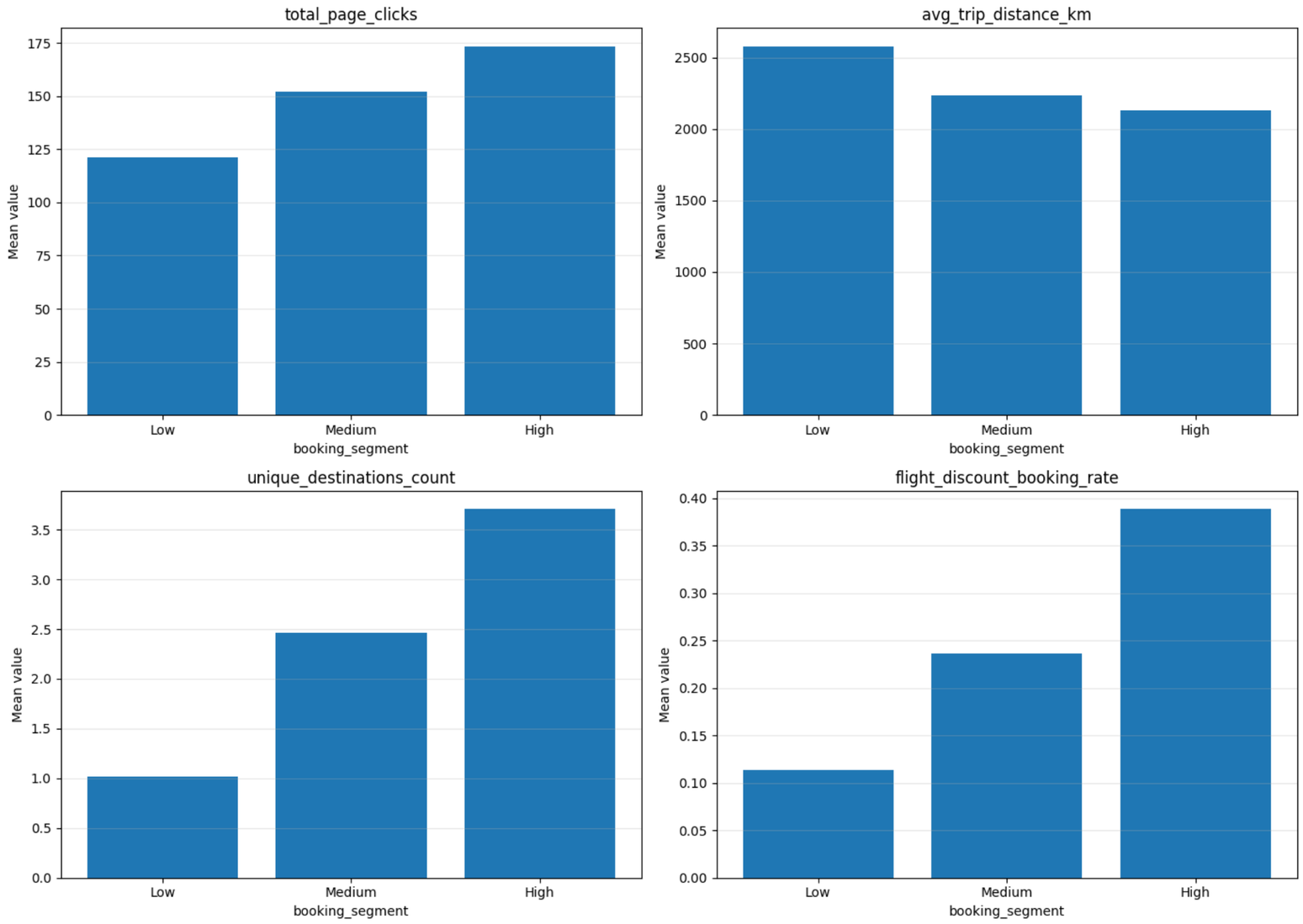
Cancellations were observed only in bundled flight + hotel bookings (4.87%).

Most sessions did not convert, but converted sessions often included both flight and hotel bookings.

These session-level patterns were then aggregated into a user table with 30+ engineered behavioral features.

Key User-Level Signals for Perk Assignment

Derived from aggregated session behavior through user-level feature engineering.



Higher Engagement

High-booking users generated the most page clicks.

Greater Travel Variety

Unique destination count increased strongly from Low to High.

Stronger Discount Responsiveness

Discount-related booking rates increased across the groups.

Distance Mattered Less

Average trip distance was not a major differentiator.

At the user level, the clearest differentiators were engagement, travel variety, and discount responsiveness.

A Decision-Oriented Approach to Perk Assignment

Available Perks

Start with the five rewards TravelTide can actually offer.

Relevant Behaviors

Define the customer behaviors that make each perk valuable.

Perk-Specific Feature Sets

Select the most relevant user-level features for each perk.

Scoring Input

These feature sets become the basis for the scoring logic shown next.

Example Perk-Specific Feature Sets

Free Hotel Meal

hotel-oriented behavior

- hotel_booking_rate
- avg_nights
- completed_hotel_bookings

Free Checked Bag

baggage-heavy flight behavior

- flight_booking_rate
- avg_checked_bags
- return_flight_booking_rate

Rather than clustering users first, we began with the five available perks and selected the most relevant user-level features for each reward.

Engineering the User Profile

A transparent scoring framework for perk assignment

1. User-Level Features

- [%] Hotel Booking Rate (%)
- [#] Avg. Checked Bags (#)
- [%] Flight Discount Booking Rate (%)
- [%] Cancellation Rate (%)
- [%] Hotel Attach Rate (%)
- [#] Avg. Nights (#)

2. Percentile Rank Transformation

Uniform Scale (0.0 – 1.0)

Makes different user-level metrics directly comparable on a 0–1 scale.

3. Perk-Fit Scores

- Free Checked Bag Score
- Free Hotel Meal Score
- Exclusive Discounts Score
- No Cancellation Fees Score
- Free Hotel Night with Flight Score

4. Final Assignment

Highest Score = Final Assigned Perk

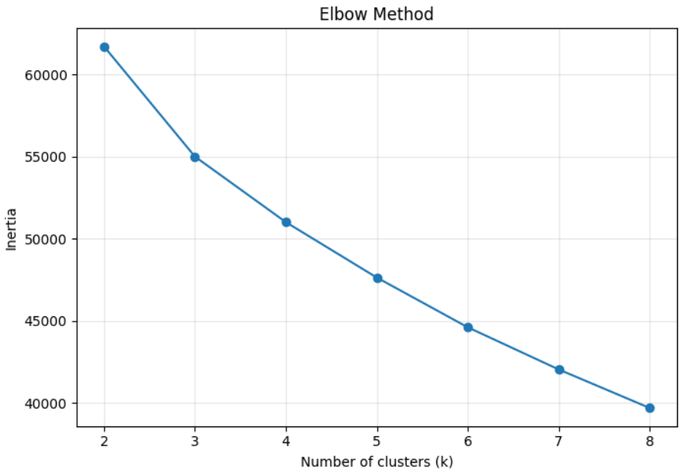
Each user is assigned to the perk with the highest overall fit score. This creates one mutually exclusive final segment per user.

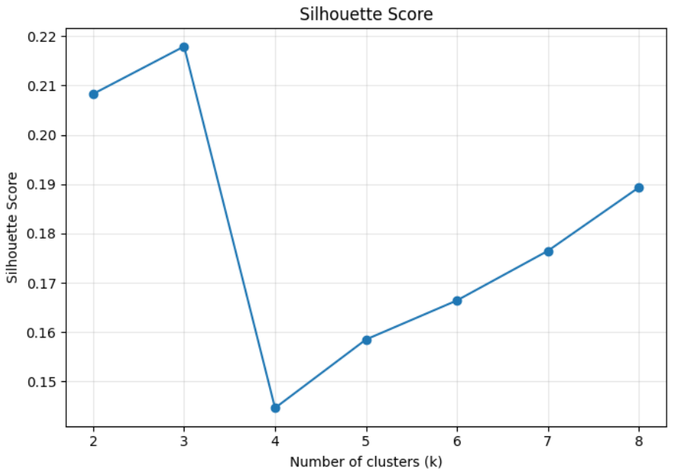
Selected user-level features were transformed into percentile ranks and then combined into perk-specific fit scores.

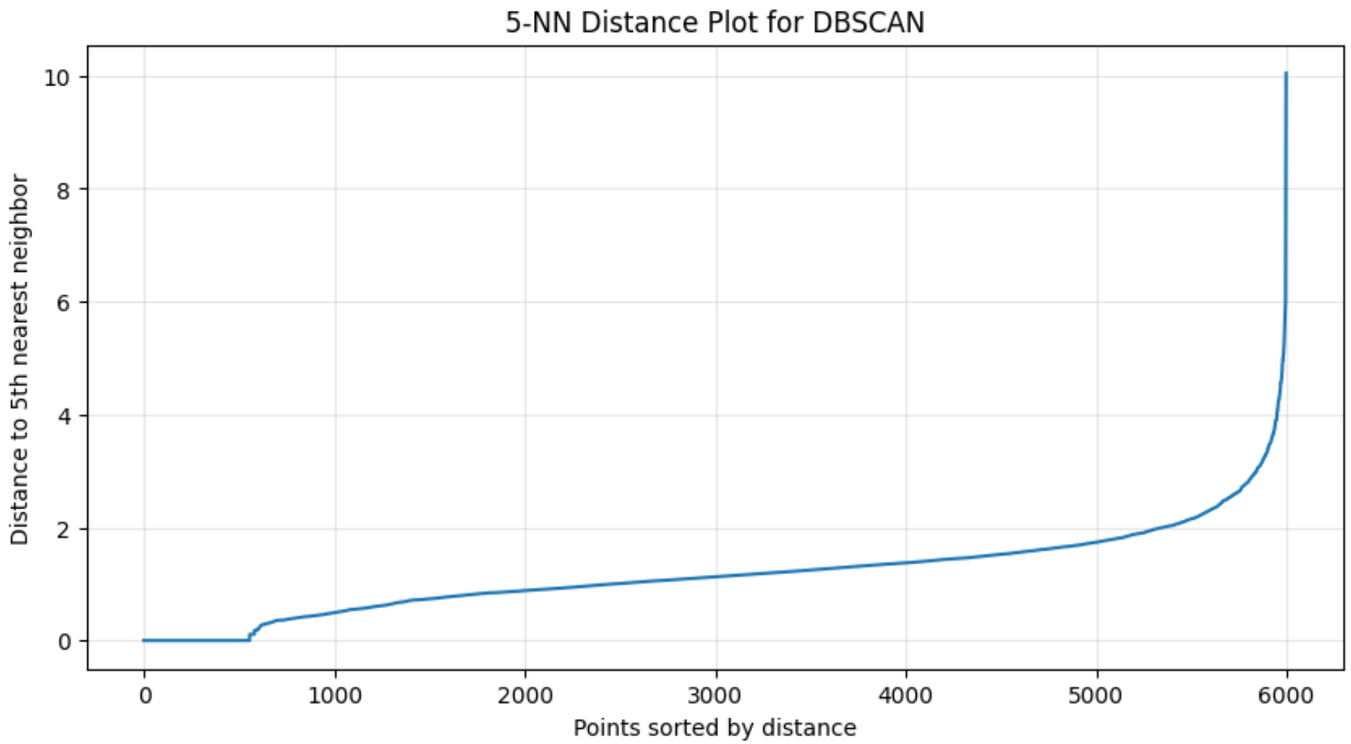
Final Customer Segments and Their Assigned Perks

Each eligible customer is assigned to one segment based on highest behavioral fit.

Evaluating Segmentation Methodologies: ML vs. Rule-Based







K-Means

Observation
K-Means indicated only moderate clustering structure, with the clearest signal around k = 3.

Insight
User behaviors remained overlapping rather than sharply separated into natural segments, limiting ML's usefulness for direct perk assignment.

DBSCAN

Check
The 5-NN distance plot suggested one dominant dense region with edge-case users, making DBSCAN more useful for outlier detection than for stable business segmentation.

Takeaway
Rule-based segmentation remained the preferred approach because it offered stronger business alignment, clearer interpretability, and more stable user-to-perk assignment for deployment.

Strategic Recommendations and Immediate Next Steps

Use the 5 finalized rule-based segments as the primary targeting framework for TravelTide's upcoming personalized email campaign.

Launch segmented emails and compare open and sign-up rates against a non-segmented control group to measure incremental ROI.

Continue using exploratory ML in the background to monitor shifting user behavior and detect emerging anomalies.

Recommended success metrics: email open rate, perk sign-up rate, and incremental ROI versus a generic control campaign.